## Dr. Oun Al-iedani

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### Qualifications

- **2014 – 2019** PhD (Magnetic Resonance In Medicine), University of Newcastle, Australia.

### Present Employment

- **2025 – present** Postdoctoral Researcher, MS Fellow, School of Biomedical Sciences and Pharmacy, University of Newcastle, Australia.
- **2023 – 2024** Postdoctoral Researcher, School of Biomedical Sciences and Pharmacy, University of Newcastle, Australia.
- **2022 – 2024** Postdoctoral Researcher, MS Clinic Research, John Hunter Hospital (JHH), Australia.
- **Oct 2022 – Feb 2023** Imaging Data Analyst, Brain Neuromodulation Research Program & Active Living Research Program, School of Psychological Sciences, University of Newcastle, Australia.
- **Nov 2022 – Feb 2023** Imaging Data Analyst, IHR Program, School of Biomedical Sciences and Pharmacy, University of Newcastle, Australia.
- **2020 – 2022** Postdoctoral Researcher, Advanced MRI technology on Multiple Sclerosis, School of Health Sciences, University of Newcastle, Australia.

### Society and Memberships

- **2016 – present** Member, International Society for Magnetic Resonance in Medicine (ISMRM).
- **2017 – present** Member, Australian & New Zealand Society of Magnetic Resonance (ANZMAG).
- **2016** Member, Australian Society of Medical Research (ASMR).

### Current Research: Supervision and Leadership

As an Early Career Researcher (ECR), Al-iedani has demonstrated strong leadership and mentorship capabilities. He has supervised and provided guidance to nine University of Newcastle PhD and Honours students, fostering the next generation of neuroimaging researchers.

Al-iedani has managed MS projects for five years and also applied artificial intelligence algorithms to differentiate between brain structures for easier recognition of brain pathology. As a contribution to the clinical program in MS, Al-iedani has combined his experience in the analysis of MRI, MR Spectroscopy and Diffusion MRI with new advanced techniques involving learning algorithms.

Furthermore, Al-iedani has applied his expertise in advanced MRI techniques beyond MS, notably evaluating and validating the application of Amide Proton Transfer-weighted (APTw) imaging for brain cancer (grade 4 gliomas), demonstrating his versatility and impact across neurological disorders.

Currently, he is leading an ongoing 10-year follow-up project utilising Multi-modal MRI for personalised MS progression predictionis (MS).

## Professional Service

- **Editorial Board Member:** BMC Medical Imaging journal.
- **Reviewer:** Multiple Sclerosis and Related Disorders; Brain Research Bulletin; Neural Plasticity; ISMRM (2021, 2022, 2023, 2024).

## Awards and Honours

- **2025** MS Australia Postdoctoral Fellowship
- **2025** IHRP EMCR Travel Award
- **2024** IHRP EMCR Travel Award
- **2021** ISMRM Magna Cum Laude Merit Award
- **2020** ISMRM Education Stipend
- **2019** ISMRM Magna Cum Laude Merit Award
- **2017** ISMAR Travel Grant (ANZMAG Conference)

## Grants and Funding

- **2024:** MS Australia, Hunter New England Local Health District and HMRI $434,316(CIA). Project: "Investigate Multi-modal MRI for personalised MS progression prediction."
- **2023:** Mark Hughes Foundation Research Innovation Grant. $120,000 (CIA). Project: "Evaluate APTw imaging for brain cancer."
- **2023:** Hunter Medical Research Institute grant. >$25,000 (CIB). Project: "Investigating the impact of diet on MS using novel MRI approaches."
- **2020:** Hunter Medical Research Institute grant. $50,000 (CIB). Project: "Mindfulness and exercise-based approaches for MS fatigue."
- **2020:** Hunter New England Local Health District. Co-investigator (CI-D). Project: "Effect of dimethyl fumarate (Tecfidera) on microstructure and biochemistry in the brain of MS patients (TECSPEC)."
- **2022:** HMRI Immune Health Program Seed Funding.
- **2021:** CBMHR Seed Funding Grant Round, University of Newcastle.
- **2018:** Grant scholarship, University of Newcastle.
- **2018:** University of Newcastle International Postgraduate Research Scholarship.

## Achievements & Impact

Dr Al-iedani has published 29 peer-reviewed articles in high-impact journals, including the *Journal of Biomedical Science*, *Multiple Sclerosis and Related Disorders*, and *NMR in Biomedicine*. His work has garnered over 300 citations, resulting in an h-index of 11.

His most recent publications focus on the cutting-edge application of diffusion MRI and tractography to improve early MS diagnosis and monitor disease progression.

His research has contributed to a five-year longitudinal MS study examining treatment effects on clinical and neurochemical outcomes, applied AI tools for brain structure differentiation, and developed MRI signatures predictive of cognitive decline in MS. He has over 40 peer-reviewed conference abstracts, with 25 published in proceedings.